📊 Executive Summary – Telco Customer Churn Analysis

This project provides a comprehensive analysis of the **Telco Customer Churn dataset**, focusing on identifying the primary factors influencing customer attrition. Through statistical exploration and visualization, the study highlights churn drivers and offers actionable recommendations for improving customer retention.

🔑 Key Findings

- **Overall Churn Rate: ~26.5%** of customers discontinued services, indicating a significant retention challenge.
- **Tenure Impact:**
    - Customers with tenure **<12 months** represent **~50% of churn cases**.
    - Customers with tenure **>36 months** show churn rates below **10%**, demonstrating loyalty growth over time.
- **Contract Type:**
    - **Month-to-month contracts: ~43% churn rate**.
    - **One-year contracts: ~11% churn rate**.
    - **Two-year contracts: ~3% churn rate**.
- **Monthly Charges:**
    - Customers paying **>$70/month** → churn **~40%**.
    - Customers paying **<$40/month** → churn **~15%**.
- **Payment Method:**
    - **Electronic check users:** churn **~45%**.
    - **Credit card/Bank transfer users:** churn **~15–20%**.
- **Service Usage:**
    - Multi-service customers churn **20% less** than single-service users.

📈 Visual Insights

- **Bar charts** highlight churn concentration among month-to-month contract holders.
- **Heatmaps** reveal strong correlation between **monthly charges, tenure, and churn probability**.
- **Distribution plots** show churn risk is highest in the first year and declines steadily with tenure.

🎯 Business Implications

- **Retention Strategy:** Transition customers from month-to-month to longer contracts → potential churn reduction of **40%**.
- **Pricing Optimization:** Re-evaluate high-charge plans → churn reduction of **20–25%**.

- **Payment Incentives:** Encourage secure payment methods → churn reduction of **25%**.
- **Early Engagement:** Focus loyalty programs on new customers (<12 months tenure) → prevent early exits.

📌 Key KPIs

Metric	Value
Churn Rate	26.5%
Retention Rate	73.5%
High-Risk Group	Month-to-month + electronic check users (~45% churn)
Low-Risk Group	Two-year contract holders (~3% churn)

📑 Conclusion

The analysis demonstrates that churn is heavily influenced by **contract type, tenure, monthly charges, and payment methods**. By implementing targeted retention strategies—such as promoting long-term contracts, optimizing pricing, and incentivizing secure payment methods—Telco companies can significantly reduce churn and improve customer lifetime value.